

claude-wsl-ubuntu / 4f364eb7-a600-4a2c-9b37-e5a313b6db9d

Metadata

```
- Source: `claude-wsl-ubuntu`  
- Kind: `claude`  
- Source file: `\\wsl.localhost\Ubuntu\home\avidullu\.claude\projects\home-avidullu-projects-khelsutra-guru\4f364eb7-a600-4a2c-9b37-e5a313b6db9d\subagents\workflows\wf_fa7a2077-3a9\agent-adb6a4b6311cd61d7.jsonl`  
- SHA-256: `e254a94f5ef230f1d18d71845a2d136d013fc560b4ce78aead0f3f193254e57e`  
- Source modified: `2026-06-29T10:54:00+00:00`  
- Imported at: `2026-07-05T16:48:30+00:00`  
- project: `wf_fa7a2077-3a9`  
- session_id: `4f364eb7-a600-4a2c-9b37-e5a313b6db9d`
```

Transcript

1. user (2026-06-29T10:52:26.015Z)

You are assembling a single authoritative P2b runbook from these structured doc/code digests of the badminton-highlight-indexer rally-detection work.

Digests (JSON):

```
[  
  {  
    "key": "fixes-plan",  
    "data": {  
      "p2b_current_status": "P2b is NOT done ? it is the only open/blocked row in the §7 tracker. Verbatim row: `\"| P2b | **Promote P0b** : golden-set A/B at SERVED windowing ? flip flag default-ON if no regression | P2a | ? owner | ? | ? |\". Status glyph = ? (Todo), Gated? = \"? owner\", PR = \"?\" (none). It depends on P2a, which is ? done (\"this PR\"). The §9 Definition-of-Done item for it is unchecked: `\"[ ] **P2 promotion** : golden-set A/B at SERVED windowing, then flip the flag default-ON if no regression. *(owner-gated, real video ? see §8 Q2)*\". It is owner-gated because it needs the golden corpus + GPU/WSL-extracted trajectory features that do not exist offline, and finding B is a no-op under dense tracking so the A/B only shows signal on clips where the shuttle tracker drops out (cannot be measured on synthetic/offline fixtures).\", \"ab_runbook_commands\": [  
        \"python -m backend.eval.rally_seg_eval \"\",  
        \" --manifest <golden_manifest.json> --features-dir <golden_features_dir> \"\",  
        \" --preset served --vs served --vs-inactivity-temporal-density\"  
      ],  
      "decision_rule": "From §8 Q2 / §9, verbatim: base = SERVED with the flag OFF, candidate = SERVED with it ON; the run `diffs tolF1 at the **SERVED** operating point`. Decision: `***Promote** by flipping `IndexingConfig.inactivity_temporal_density` to `True` (a one-line PR) **only if tolF1 does not regress**.` §9 restates: `flip the flag default-ON if no regression.` The headline gate is tolF1-does-not-regress at the SERVED operating point. The doc does NOT specify a numeric regression epsilon, p-value, or CI rule for this gate ? those are absent from the doc (see secondary_metrics).\",  
      "headline_metric": "tolF1 (tol_f1) ? the R2 tolerance-match F1 at the SERVED operating point. This is the only headline metric named in the §8 Q2 / §9 decision rule; the rule is `promote only if tolF1 does not regress.`\",  
      "promote_steps": [  
        \"Obtain golden corpus + GPU/WSL-extracted trajectory features (owner-only; do not exist offline).\",  
        \"Run the §8 Q2 A/B command: python -m backend.eval.rally_seg_eval --manifest <golden_manifest.json> --features-dir <golden_features_dir> --preset served --vs served --vs-inactivity-temporal-density (base=SERVED flag-OFF, candidate=SERVED flag-ON).\",  
        \"Check the headline gate: confirm tolF1 does not regress at the SERVED operating point.\",  
        \"If no regression: flip IndexingConfig.inactivity_temporal_density from False to True in backend/config/models.py (line 199) ? a one-line PR. This is the exact one-line change to promote.\",  
        \"Update §7 P2b row status ??? with the promotion PR link, and check the §9 P2-promotion
```

```

box."
],
"reject_steps": [
    "The doc gives no explicit reject procedure. Implicit from the decision rule: if tolF1
    regresses at the SERVED operating point, do NOT flip the flag ? leave
    inactivity_temporal_density default-OFF (the merged state is already a zero-regression no-op in
    production since P0b shipped flag-OFF). No code change is needed to reject; P0b stays gated
    default-OFF and P2b remains ? (or is closed as rejected with a note).",
],
"doc_update_steps": [
    "After promotion, update $7 tracker P2b row: change Status ??? and PR '?'<promotion PR
    link> (the $7 tracker is the source of truth).",
    "Check the $9 Definition-of-Done P2-promotion checkbox: '[ ] **P2 promotion**: golden-
    set A/B at SERVED windowing, then flip the flag default-ON if no regression.'",
    "Update the doc header 'Last updated' date and, when all rows land, flip the Status
    header from IN PROGRESS toward DONE per the DRAFT ? IN PROGRESS ? DONE ? archived lifecycle.",
    "Note: $1 'Tracking anchors' says $7 is the source of truth and is indexed in
    docs/README.md; the plan Extends docs/RALLY_DETECTION_QUALITY_REPORT.md.",
    "Separately track the still-open sibling-repo item in $9: '[ ] Warning log added for
    malformed rows in rally_cvat.py. (sibling-repo PR, separate)' ? though $7 marks P1 as ? done via
    sdc#50, creating an inconsistency between $7 (P1 ?) and $9 (rally_cvat warning unchecked).",
],
"served_regress_eps": "ABSENT from the plan doc. The doc never names 'served_regress_eps';
its decision rule is the qualitative 'tolF1 does not regress'. Grounded in code (not the doc):
backend/eval/promotion.py:126 and backend/eval/serve_contrast.py:112 define served_regress_eps
with default 0.0, meaning '0.0 = no served-mean drop allowed' (promotion.py:147). That
promotion-gate eps default of 0.0 is the closest concrete value, but it is NOT referenced by
RALLY_DETECTION_FIXES_PLAN.md.",
"secondary_metrics": [
    "ABSENT from RALLY_DETECTION_FIXES_PLAN.md ? the requested secondary metrics |?end|,
    rally-end precision, paired sign-test, and CI delta are NOT mentioned anywhere in the plan doc.
    The doc's $8 Q2 / $9 decision rule cites only tolF1.",
    "These metrics DO exist in the eval harness code (not the doc):
    backend/eval/rally_seg_eval.py emits |?end| in its per-video table (line ~523: '| ... | |?start|
    | |?end| | split | merge |') and computes a paired sign-test + CI / per-tier delta (module
    docstring line 13: 'diff two windowings with a paired sign-test + CI (the per-tier ``delta``)';
    line ~358 'sign-test ... on both tIoU-F1 and the R2 tol_f1'; aggregate CI via _ci(a['tol_f1'])
    at line ~534). 'rally-end precision' as a named metric does not appear; the harness reports
    tolP/tolR/tol_precision (tolerance precision/recall), not a metric literally called 'rally-end
    precision'.",
    "Conclusion: treat |?end|, paired sign-test, and CI-delta as harness outputs you would
    inspect when running the A/B, but the PLAN DOC does not elevate any of them into the
    promote/reject rule.",
],
"section7_tracker": "$7 'Deliverables & progress tracker ? source of truth'. Legend: ?
Todo . ? In progress . ? Done . ? Blocked/gated. One small PR per row, branched from
origin/master, no stacking. Rows verbatim: (1) '| P0a | Fix `_spread` zero-collapse (behaviour-
neutral, unconditional) | ? | No | ? | #394 |'. (2) '| P0b | `_trim_inactive_tail` temporal
density behind `inactivity_temporal_density` flag (default-OFF) | ? | No | ? | #394 |'. (3) '|
P1 | Warn on rows silently skipped by `load_canonical_csv` (sibling `sports-data-collector`) | ?
| No | ? | sdc#50 |'. (4) '| P2a | Expose the flag in the `rally_seg_eval` A/B CLI
(`--[vs-]inactivity-temporal-density`) so the promotion run is one command | P0b | No | ? | this
PR |'. (5) '| P2b | **Promote P0b**: golden-set A/B at SERVED windowing ? flip flag default-ON
if no regression | P2a | ? owner | ? | ? |'. SUMMARY: P0a=DONE(#394), P0b=DONE(#394),
P1=DONE(sdc#50), P2a=DONE(this PR), P2b=TODO/owner-gated(no PR). NOTE: P0a and P0b both list PR
#394, though the $0/header narrative says P0 is 'implemented + unit-tested in this same PR
(#393)' ? header references #393 while the $7 table references #394 (an internal inconsistency
in the doc).",
"paths_referenced": [
    "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
    indexer/docs/RALLY_DETECTION_FIXES_PLAN.md",
    "backend/pipeline/detectors/rally_gate.py (P0a _spread fix; $4 + $10 code anchor)",
    "backend/pipeline/detectors/tracknet_runner.py (P0b _trim_inactive_tail
    temporal_density; $4 + $10 code anchor; flag at line 329/426 confirmed in code)",
    "src/parsers/cvat/rally_cvat.py (P1 load_canonical_csv warning; sibling sports-data-

```

```

collector repo; $4 + $10)",
    "backend/config/models.py ? IndexingConfig.inactivity_temporal_density (the EXACT one-
line promote flag; verified at line 199, default False)",
    "backend/eval/rally_seg_eval.py ? the rally_seg_eval A/B CLI run via `python -m
backend.eval.rally_seg_eval` (P2a / $8 Q2); flags --inactivity-temporal-density and --vs-
inactivity-temporal-density (verified lines 587/603/612)",
    "backend/eval/windowing.py ? WindowingPreset / SERVED preset plumbing (config ? eval
preset; verified)",
    "backend/pipeline/segmenters/trajectory_hybrid.py ? trajectory_hybrid plumbing config ?
segmenter (verified line 250/305)",
    "<golden_manifest.json> ? golden manifest, a --manifest placeholder in $8 Q2 (no
concrete path given in doc)",
    "<golden_features_dir> ? golden features dir, a --features-dir placeholder in $8 Q2 (no
concrete path given in doc)",
    "docs/RALLY_DETECTION_QUALITY_REPORT.md (doc this plan Extends)",
    "docs/RALLY_DETECTION_CODE_REVIEW.md ($10 internal docs reference)",
    "docs/README.md (where $7 tracker is indexed)",
    "backend/eval/promotion.py & backend/eval/serve_contrast.py ? where served_regress_eps
(default 0.0) actually lives; NOT referenced by the plan doc"
    ]
    },
    {
        "key": "code-review-findings",
        "data": {
            "finding_a": "{ \"id\": \"A\", \"title\": \"Subtle calculation artifact in _spread\", \"file\":
\"backend/pipeline/detectors/rally_gate.py\", \"function\": \"_spread\", \"what_was_wrong\": \"Robus
t spread trims bottom/top 5% via int()-truncated percentile indices: lo = s[max(0,
int(0.05*(len(s)-1))], hi = s[min(len(s)-1, int(0.95*(len(s)-1))]. Because int() truncates
rather than rounds, for short arrays (<20 items) the p95 index collapses to the second-to-last
element, and for len(s)==2 the spread evaluates to strictly 0.0 (zero
deadband).\", \"impact\": \"Not a crash. Currently benign because the play axis is estimated over
the entire video's thousands of trajectory points. Risk is latent: if reused on small window
slices, spread silently collapses to zero.\", \"fix\": \"_spread now rounds the p95 index UP with
math.ceil(), so a 2-element (or any short) input falls back to the full min..max range instead
of collapsing to 0.0. On large arrays ceil-vs-truncate differ by at most one sample, so the
robust p5/p95 trim is unchanged in practice.\", \"behavior_change\": \"None. Behaviour-neutral,
lands unconditionally (not flag-gated).\", \"status\": \"Done\", \"pr\": \"#394 (P0a). Note: review
doc front-matter says PR #393; the fixes-plan progress tracker (source of truth) lists
#394.\", \"definition_of_done\": \"Bug fixed and tested; behaviour-neutral, unconditional.\"}",
            "finding_b": "{ \"id\": \"B\", \"title\": \"Sensitivity to sparse tracking in
_trim_inactive_tail (rally-end / inactivity temporal-density issue)\", \"file\": \"backend/pipeline
e/detectors/tracker_runner.py\", \"function\": \"_trim_inactive_tail\", \"heuristic\": \"R4
(Inactive Tail Trimming) ? trims a candidate window's end back to the last point where a
trailing trail_frames window has sufficient density of fast frames (velocity >
rally_thresh).\", \"what_was_wrong\": \"The density check is fast_sum / win_count >= min_density,
but win_count is the number of VISIBLE, ACTIVE frames (samples) in the trailing boundary, NOT
total elapsed time. If the detector loses the shuttle for ~1.8s and picks up only 1 fast frame
at the window's end, win_count is tiny (e.g. 1-2), so fast_sum/win_count can evaluate to 1.0
(100% density). Sparse trailing points then defeat the trimmer, leaving dead-time uncut and
artificially inflating the candidate window's END boundary. Partially mitigated by merge_gap
breaking trajectories, but remains a boundary-logic quirk.\", \"what_the_fix_changes\": \"Adds an
inactivity_temporal_density flag. When ON, the density DENOMINATOR becomes the trailing window's
TEMPORAL length, min(trail_frames, frames[i] - frames[0] + 1), instead of win_count (the count
of active samples that sparse tracking inflates). Resolved as STRICT temporal density (no
separate few-frames fallback ? Q1 resolved).\", \"why_noop_under_dense_tracking\": \"The 'group'
holds only active frames, so under dense tracking the temporal denominator min(trail_frames,
frames[i]-frames[0]+1) equals the sample-count denominator win_count. The two denominators are
identical, so flipping the flag is a no-op on well-tracked clips and changes behaviour ONLY in
the sparse-tracking case. Corollary: signal appears only on clips where the shuttle tracker
actually drops out ? so it cannot be measured on synthetic/offline
fixtures.\", \"why_flag_gated\": \"R4 is DEFAULT-ON in production (window_inactivity_end = 1.5,
promoted at the R1 milestone). An earlier draft wrongly called R4 default-OFF; corrected
2026-06-29. Because R4 is live, the fix is a REAL behaviour change on sparse footage, so it
ships behind a default-OFF flag to keep the merge a zero-regression no-op in prod until measured

```

on the golden set.\", \"flag\": {\"name\": \"inactivity_temporal_density\", \"default\": \"OFF (False)\", \"plumbing\": \"config (IndexingConfig.inactivity_temporal_density) -> trajectory_hybrid segmenter -> eval WindowingPreset; also exposed in rally_seg_eval A/B CLI as --[vs-]inactivity-temporal-density (P2a).\"}, \"status\": \"Implemented + unit-tested in both flag states. Done as P0b. Promotion to default-ON (P2b) is owner-gated and NOT done.\", \"pr\": \"#394 (P0b); CLI toggle P2a marked 'this PR'. (Review doc front-matter cites #393; fixes-plan tracker is source of truth at #394).\"}],

\"finding_b_acceptance\": {\"summary\": \"Promotion (P2b) is owner-gated: requires golden corpus + GPU/WSL-extracted trajectory features that do not exist offline.\", \"measurement_command\": \"python -m backend.eval.rally_seg_eval --manifest <golden_manifest.json> --features-dir <golden_features_dir> --preset served --vs served --vs-inactivity-temporal-density\", \"ab_setup\": \"base = SERVED windowing with flag OFF; candidate = SERVED windowing with flag ON. Measured at the SERVED operating point specifically because R4 is default-ON in prod, so this captures B's real end-boundary effect on sparse footage.\", \"metric\": \"tolF1 (tolerant F1) at the SERVED operating point.\", \"acceptance_criterion\": \"Promote (flip IndexingConfig.inactivity_temporal_density to True, a one-line PR) ONLY IF tolF1 does not regress (no regression at SERVED).\", \"measurability_caveat\": \"B is a no-op under dense tracking, so the A/B only shows signal on clips where the shuttle tracker actually drops out; cannot be measured on synthetic/offline fixtures ? hence owner-gated on real labelled video.\", \"gate_status\": \"P2b = Todo, Blocked/gated (owner). Offline unit tests pass in both flag states; golden-corpus regression run is the outstanding owner-gated step.\"}],

\"finding_c\": {\"id\": \"C\", \"title\": \"Graceful degradation on incomplete CSV validation\", \"file\": \"src/parsers/cvat/rally_cvat.py (sibling sports-data-collector repo)\", \"functions\": \"parse_cvat_boxes / load_canonical_csv\", \"what_was_wrong\": \"parse_cvat_boxes searches zips for an .xml ending in 'annotations.xml' or else ANY .xml. In load_canonical_csv, safe_int() guards malformed rally_number fields; a fully malformed row (that doesn't break CSV parsing) is silently 'continue'd. Stable, but a severely malformed CSV could silently drop half its rallies without failing the import gate ? an observability gap.\", \"fix\": \"Add a logger.warning before each 'continue' in load_canonical_csv so skipped/malformed rows are surfaced during ingestion. Does not change parsing behaviour, only adds observability.\", \"scope_note\": \"Lives in the sibling sports-data-collector repo, so it ships as a SEPARATE PR (cannot be in the in-repo #393/#394).\", \"status\": \"Done in sibling repo as P1, PR sdc#50. (Definition-of-done checkbox in fixes plan still shows the warning-log item unchecked, an inconsistency vs the P1 tracker row marked Done).\"}],

},
{
\"key\": \"code-anchors\",
\"data\": {

\"trim_inactive_tail\": \"ANCHOR 1 ? backend/pipeline/detectors/tracknet_runner.py ::
_trim_inactive_tail (def at line 450; body 450-497).\\n\\nHOW THE R4 RALLY-END TRIM WORKS (lines 450-497):\\n- Signature (450-456): _trim_inactive_tail(group: List[Tuple[int,float]], trail_frames: int, min_density: float, rally_thresh: float, temporal_density: bool = False) -> List[Tuple[int,float]].\\n- Purpose (docstring 457-474): after a rally the shuttle keeps drifting slowly (picked up / walked / knock-up) above velocity_thresh, so velocity windowing OVER-EXTENDS the END. A frame is \\\"fast\\\" (real rally motion) iff velocity > rally_thresh (a HIGHER bar than the active threshold). The rally is \\\"on\\\" at frame i while its trailing trail_frames window holds >= min_density fraction of fast frames; it trims back to the LAST such \\\"dense\\\" frame ? everything after is sustained slow drift = rally over. ONLY CUTS (recall cannot rise). Fail-safe: a window whose tail never goes slow is untouched, and if no fast-dense point exists at all the group is kept whole (no over-trim). O(n) two-pointer; the START is left alone.\\n- Guard (475-476): if trail_frames <= 0 or len(group) < 2: return group unchanged.\\n- Implementation (477-497): frames = [f for f, _ in group] (478); fast = [1 if v > rally_thresh else 0 ...] (478); two-pointer sliding window with lo, fast_sum, last_dense=-1 (479-481); for each i: fast_sum += fast[i] (483), slide lo forward while frames[lo] <= frames[i] - trail_frames subtracting fast[lo] (484-486). Density check (487-495): chooses the DENOMINATOR per the gate, then if denom > 0 and fast_sum/denom >= min_density: last_dense = i (495-496). Returns group[: last_dense + 1] if last_dense >= 0 else group (497).\\n\\nHOW IT IS GATED ON temporal_density (lines 467-474 docstring + 487-494 logic):\\n- The temporal_density flag selects only the density DENOMINATOR (code-review \\\"finding B\\\"). group carries ONLY active frames.\\n- temporal_density == True (487-490): denom = min(trail_frames, frames[i] - frames[0] + 1) ? the trailing window's TEMPORAL length (full trail_frames once that much footage elapsed, else footage-so-far); sparse tracking cannot inflate density.\\n- temporal_density == False (legacy, 491-494): denom = (i - lo + 1) ?

the COUNT of active samples in the trailing window; a lone fast sample after a long tracking gap reads 100% dense and holds the tail open.\n- Under DENSE tracking the two denominators are identical, so the flag is a no-op on well-tracked clips. Per docstring (473-474) it ships flag-gated default-OFF (default False = legacy/promoted behaviour) while R4 itself is default-ON in prod.\n\nCALL SITE (lines 413-429): R4 runs inside trajectory_to_action_windows, AFTER the cap split (_split_overlong_groups, 405-411) and BEFORE the R5 blob fallback (431-439). Gated on window_inactivity_end > 0 (418): tw = int(window_inactivity_end * fps) (419); maps each group through _trim_inactive_tail(g, tw, inactivity_min_density, inactivity_rally_thresh, temporal_density=inactivity_temporal_density) (421-427). The runner param inactivity_temporal_density: bool = False is declared at line 329.",

"config_flag_state": "ANCHOR 2 ? backend/config/models.py :: field inactivity_temporal_density.\n\nEXACT CURRENT DEFAULT (verbatim, line 199):\n inactivity_temporal_density: bool = False\n\n- LINE: 199.\n- DEFAULT VALUE: False (bool).\n- SURROUNDING CONFIG CLASS: class IndexingConfig(BaseModel) (declared at line 146).\n- DOC COMMENT (lines 193-198): \"R4 density DENOMINATOR fix (code-review finding B), default OFF until measured at the SERVED operating point. The R4 trim's fast-fraction denominator is the COUNT of active samples in the trailing window, which sparse tracking inflates (one fast sample after a gap -> 100% dense -> tail held open). When True, the denominator is the window's TEMPORAL length instead. No-op under dense tracking (sample-count == temporal-span), so this can flip with zero regression on well-tracked clips; flag-gated default-OFF since R4 itself is default-ON.\" \n\n- RELATED IndexingConfig R4 fields nearby: window_inactivity_end: float = 1.5 (line 190, R4 trim default-ON), inactivity_rally_thresh: float = 0.20 (191), inactivity_min_density: float = 0.30 (192). (Note: the eval WindowingPreset defaults for rally_thresh/min_density differ ? 0.08/0.34 ? those are separate defaults in windowing.py; the SERVED preset overrides them from this config.)",

"rally_seg_eval_cli": "ANCHOR 3 ? backend/eval/rally_seg_eval.py :: _build_preset (def 574, body 574-621) + the two temporal-density CLI flags.\n\n_build_preset (574-621): resolves the base (--preset) or cand (--vs) WindowingPreset from argparse args via dataclasses.replace overrides for cap / hard-cap / blob-fallback / inactivity. cand falls back to the base override when its --vs-* variant is unset (so a baseline-vs-candidate promotion check runs in one shot) ? docstring 575-578.\n\n- side == \"base\" (581-587): preset = _resolve_preset(args.preset); reads cap, hard_cap, blob_fallback, inactivity_end, and temporal_density = args.inactivity_temporal_density (587).\n\n- side == \"cand\" (588-604): preset = _resolve_preset(args.vs); each knob falls back to the base value when its --vs-* is unset. For temporal_density (602-604): temporal_density = args.vs_inactivity_temporal_density or args.inactivity_temporal_density ? i.e. --vs gets temporal-density if EITHER its own --vs flag OR the base --inactivity-temporal-density flag is set.\n\n- Override application (605-620): if temporal_density: preset = replace(preset, inactivity_temporal_density=True) (611-612). (Plus cap 605-606, hard_cap 607-608, blob_fallback 609-610, blob_factor 613-614, inactivity_end 615-616, inactivity_rally_thresh 617-618, inactivity_min_density 619-620.) Returns the preset (621).\n\n\nCLI FLAGS (defined in main(), def at line 624):\n\n- --inactivity-temporal-density (712-717): action=\"store_true\"; help \"R4 density over the trailing window's TEMPORAL length, not the active-sample count, on --preset (finding-B fix; no-op under dense tracking, trims sparse-track tails)\". Stored as args.inactivity_temporal_density, read at 587 (base) and 603 (cand fallback).\n\n- --vs-inactivity-temporal-density (718-723): action=\"store_true\"; help \"R4 temporal-density on --vs (default: same as --inactivity-temporal-density); use '--preset served --vs served --vs-inactivity-temporal-density' for the P2 promotion A/B\". Stored as args.vs_inactivity_temporal_density, read at 603.\n\n\nHOW THE A/B IS WIRED: main() builds base = _build_preset(args, \"base\") (783) and cand = _build_preset(args, \"cand\") (786). The canonical promotion A/B is '--preset served --vs served --vs-inactivity-temporal-density' (P2 promotion A/B per the 722 help): both sides resolve to the served preset, but only the --vs side flips inactivity_temporal_density=True, isolating the finding-B denominator change at the served operating point. The store_true flags default to False, so absent flags reproduce served/legacy behaviour on both sides.",

"windowing_preset": "ANCHOR 4 ? backend/eval/windowing.py :: how WindowingPreset carries the flag.\n\n\nWindowingPreset is a @dataclass(frozen=True) (decorator line 48, class at 49).\n\n- FIELD (lines 74-77): doc comment 74-76 (\"R4 density-denominator fix (finding B), default OFF. When True the R4 fast-fraction is taken over the trailing window's TEMPORAL length, not the count of active samples (which sparse tracking inflates). No-op under dense tracking; gated default-OFF until measured on SERVED.\"), then the field at line 77: inactivity_temporal_density: bool = False.\n\n- to_dict() (def 91): emits \"inactivity_temporal_density\": self.inactivity_temporal_density at line 102 (alongside inactivity_end 99, inactivity_rally_thresh 100, inactivity_min_density 101).\n\n\nHOW IT IS CARRIED INTO THE RUNNER ? build_windows (def 214): forwards the preset's flag into the TrackNetRunner windowing call. Line 237:

```

inactivity_temporal_density=preset.inactivity_temporal_density (alongside
window_inactivity_end=preset.inactivity_end 234,
inactivity_rally_thresh=preset.inactivity_rally_thresh 235,
inactivity_min_density=preset.inactivity_min_density 236). This is the bridge from the eval
preset to tracknet_runner.trim_inactive_tail's temporal_density param.\n\nHOW IT IS SOURCED
FROM CONFIG (two constructors):\n- _served_preset_from_defaults() (def 108):
inactivity_temporal_density=bool(idx.inactivity_temporal_density) at lines 137-138 (reads the
IndexingConfig attribute directly).\n- served_windowing_from_config(config, family="wasb")
(def 161): inactivity_temporal_density=bool(idx.get("inactivity_temporal_density",
SERVED.inactivity_temporal_density)) at lines 192-193 (dict-style read with the SERVED preset
value as fallback).\nSo with the current config default
(IndexingConfig.inactivity_temporal_density=False), the SERVED preset also carries
inactivity_temporal_density=False.",
    "promotion_gate": "ANCHOR 5 ? backend/eval/promotion.py :: promote_if_served_safe (def
117, body 117-224) + served_regress_eps default.\n\nserved_regress_eps DEFAULT: 0.0 ? declared
in the promote_if_served_safe signature at line 126 (served_regress_eps: float = 0.0), and
identically in served_regresses at line 42 (eps: float = 0.0). Meaning per docstring 147:
\"served regression tolerance (0.0 = 'no served-mean drop allowed')\".\n\nSERVED-SAFETY GATE
LOGIC (promote_if_served_safe, 117-224):\n- Signature (117-129): keyword-only
proposal_a/proposal_b (Optional), structural=False (123), min_n=5 (124), p_threshold=0.05 (125),
served_regress_eps=0.0 (126), n_boot=2000 (127), seed=0 (128). Returns ServedGateResult.\n- Core
rule (docstring 130-136): promote a cue ONLY if it does not regress at the SERVED windowing. The
honest bar is applied to the SERVED scores, never the proposal scores ? a proposal-only win can
never carry a promotion.\n- Step 1 (151-160): base = should_promote_for_user(served_a, served_b,
structural=..., min_n=, p_threshold=, n_boot=, seed=) ? the honest gate (min-N floor + ?F1 CI
excludes 0 + sign test) on the SERVED scores.\n- Step 2 (162-166): served =
served_regresses(served_a, served_b, eps=served_regress_eps, n_boot=, seed=); served_safe = not
served["regresses"].\n- Step 3 (168-173): optional proposal-windowing view computed only if
proposal_a/proposal_b given; reported, never promoting on its own.\n- Step 4 ? compose
(175-214): The served guard can ONLY ever REMOVE a promotion, never add one.\n *
structural==True (176-192): promote = served_safe (180), vetoed = False (181). Ships ON
regardless of lift (keep_on stays True) but only auto-promotes to served default when
served_safe; if it regresses served, promote=False while keep_on holds.\n * non-structural
(193-214): would_promote = base.promote (194). If would_promote and not served_safe ->
promote=False, vetoed=True (195-203, \"VETOED by the served-windowing guard ...\" ). Elif
would_promote -> promote=True, vetoed=False (204-210, \"promoted: clears the honest bar AND is
served-safe ...\" ). Else -> promote=False, vetoed=False, reason=base.reason (211-214).\n-
Returns ServedGateResult(promote, served_safe, vetoed_by_served=vetoed, reason,
base_decision=base, served, proposal) (216-224).\n\nserved_regresses verdict core (38-67):
regresses=True when served mean ?F1 < -eps (mean_regresses, 65) OR the served ?F1 bootstrap CI
lies entirely below -eps (ci_confidently_negative); regresses = bool(mean_regresses or
ci_confidently_negative) (67). __all__ = [\"ServedGateResult\", \"promote_if_served_safe\",
\"served_regresses\"] (35).",
    "line_refs": [
        "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/detectors/tracknet_runner.py:450-497 (_trim_inactive_tail def+body)",
        "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/detectors/tracknet_runner.py:487-494 (temporal_density denominator
gate)",
        "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/detectors/tracknet_runner.py:413-429 (R4 call site)",
        "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/pipeline/detectors/tracknet_runner.py:329 (runner
inactivity_temporal_density=False param)",
        "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/config/models.py:199 (inactivity_temporal_density: bool = False)",
        "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/config/models.py:146 (class IndexingConfig)",
        "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/config/models.py:190-192 (R4 window_inactivity_end=1.5, rally_thresh=0.20,
min_density=0.30)",
        "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/eval/rally_seg_eval.py:574-621 (_build_preset)",
        "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/eval/rally_seg_eval.py:587,602-604,611-612 (temporal_density

```

```

resolution+replace)",
    "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/eval/rally_seg_eval.py:712-723 (--inactivity-temporal-density / --vs-inactivity-
temporal-density flags)",
    "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/eval/windowing.py:48-49,74-77 (WindowingPreset dataclass + flag field)",
    "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/eval/windowing.py:237 (build_windows forwards inactivity_temporal_density)",
    "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/eval/windowing.py:137-138,192-193 (config->preset sourcing)",
    "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/eval/promotion.py:117-224 (promote_if_served_safe)",
    "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/eval/promotion.py:126 (served_regress_eps: float = 0.0)",
    "/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/backend/eval/promotion.py:38-67 (served_regresses)"
]
},
{
    "key": "mental-model",
    "data": {
        "how_it_works": "Both docs exist and were read in full.\n\n=== RALLY-DETECTION PIPELINE
MENTAL MODEL (HOW_RALLY_DETECTION_WORKS.md) ===\n\nIt is a TWO-LAYER model. Keep them mentally
separate:\n\nvideo -> frames -> [DETECTION] -> shuttle (x,y,visible) per frame -> velocity
signal -> [SEGMENTATION] -> rally windows [start,end] -> IoU vs golden labels\n\nLAYER 1 -
DETECTION ("where is the shuttle this frame?"): WASB is a CNN that, per frame, emits a heatmap
-> shuttle (x,y) + a visibility/confidence. Weights were learned by gradient descent on shuttle-
coordinate labels and are used FROZEN. A *shuttle* detector was chosen deliberately because
"shuttle in flight" is the camera-agnostic signal for "a rally is happening" - unlike
player-motion, which fooled the old heuristic on broadcast cameras that pan with players. (The
owner's videos are static tripod, so player-motion would also work, but shuttle-motion is the
robust general bet.)\n\nLAYER 2 - SEGMENTATION ("when is a rally?"): from per-frame (x,y) it
computes shuttle VELOCITY (inter-frame displacement), then THREE tunable knobs turn that into
intervals:\n- velocity_thresh: a frame is "active" only if the shuttle moves faster than this
(in flight, not lying on the floor / being picked up).\n- merge_gap: active frames within this
many seconds get joined into one rally (bridges brief occlusion gaps / the beat between two
hits).\n- min_window_duration: discard windows shorter than this (blips / false
flickers).\n\nOutput: a list of [start, end] windows = the rallies.\n\nWHERE ANNOTATION FITS (the
key point): the Roora/golden labels are rally [start,end] TIMES - they say WHEN rallies are, not
WHERE the shuttle is. So they CANNOT teach WASB's detector (that needs (x,y) labels). They CAN
grade/tune the SEGMENTATION layer: score predicted windows vs golden rallies with temporal IoU
and pick the 3 knobs that maximize it - calibrate on some rallies, measure on held-out ones so
the number is honest. Per-video calibration is the headline idea: a handful of YOUR labeled
rallies nudges the knobs to YOUR court/camera, and the IoU lift is reported on rallies it never
saw.\n\nOPEN "sparks" (not commitments): richer features than raw velocity (acceleration,
trajectory continuity through occlusion, shuttle height/arc, direction reversals ~ shots, tying
to shots_count); a tiny LEARNED segmenter instead of 3 thresholds; hysteresis (enter-high/stay-
low dual threshold); rhythm/periodicity of the hit cadence; fusion of shuttle-motion (WASB) +
person-motion (designed in MULTI_SIGNAL_FUSION_PLAN.md; the person cue is ADR-007 parked #2 /
rally_gate.py "Gate B", now built default-OFF/unmeasured in fusion_hybrid.py); and per-video
calibration as a product feature.\n\n=== NAVIGATION MAP (CODE_MAP.md) ===\n\nCold-start map for
a self-hosted video->highlight pipeline (CV/AI rally detection -> SQLite -> ffmpeg stitch) with
a segmenter-agnostic eval/calibration stack, typed Pydantic config, and per-video observability
reporting. LEGEND: @path=file, ::sym=symbol, -->dataflow, $=data contract, !=footgun/gotcha,
+=registry/extension point, WSL=runs in WSL2 conda not Windows Python. A 2026-06-10 audit-
hardening sweep changed several behaviors (UPSERT add_video, video_id-keyed detector artifacts,
real indexing.wasb.* config, timeout_sec default 1800s, WAL+busy_timeout, iou>0 match, tracked
eval_baselines/, path validation) - trust git over stale lines.\n\nArchitecture: pluggable
ABC+Registry singletons for segmenters / detectors / providers / validators / sports /
annotations. video_id = MD5 of whole file (backend/utils/hashing.py::compute_video_id, the one
helper). Typed config in backend/config/ (Pydantic v2, models.py = single source of truth for
defaults; config.json shallow-overrides and ships default_segmenter='gemini' which diverges from
the model default 'yolo_hybrid'). DB = SQLite with a PRAGMA user_version migration ladder
(currently 5).\n\nRuntime path ($1A): load config -> hash -> validator registry (Format ->

```

ResolutionFPS -> PTSContinuity -> SceneCut -> Blur -> Relevance; format produces ffprobe_meta consumed downstream) -> db.add_video (UPSERT) -> pick segmenter from registry -> seg.process_video -> [STOP: segments-only in DB; eval/regression operate here] -> emit_indexer_report in a finally (NEVER raises) -> select segments -> VideoCompiler.compile_highlights (per-clip re-encode -> concat -c copy). Async API path (#16): POST /api/process -> fast-fail -> create_job -> ThreadPoolExecutor(max_workers=1) -> 202+job_id; worker loop in backend/api/job_worker.py.\n\nSix segmenters (backend/pipeline/segmenters/): motion (pixel-diff baseline, RANDOM/fabricated metadata+events), gemini (pure-AI, no candidate phase), yolo_hybrid (person detector + AI handoff; detection extracted to detectors/person_detector.py), and three trajectory hybrids sharing trajectory_hybrid.py::TrajectoryHybridSegmenter (wasb_hybrid = LIVE shuttle substrate w/ pretrained MIT badminton weights, tracknet_hybrid, fusion_hybrid = default-OFF multi-signal). 4-phase hybrid shape: P1 chunk -> P2 trajectory->candidate windows (persisted) -> P3 pad+provider.analyze_video -> P4 add_play_segment+link.\n\nThe model-agnostic WINDOWING BRAIN is backend/pipeline/detectors/tracknet_runner.py::trajectory_to_action_windows (a @staticmethod, re-exported verbatim by WasbRunner) - this is where the 3 segmentation knobs (velocity_thresh, merge_gap, min_window_duration) actually run. WASB GPU inference is WSL-quarantined in detectors/wasb_infer.py (reboot-resumable det_raw.jsonl cache; CPU tracker always full re-run; not importable on Windows). Gate A net-crossing post-filter = detectors/rally_gate.py (pure); person-cue feature math = detectors/fusion_features.py (pure).\n\nEval/calibration stack (backend/eval/, all pure core, I/O in CLI drivers): metrics.py (the IoU/match/score spine) <- annotations.py (GT loader) / calibrate_wasb.py (load_golden_gts, make_predict_fn, BASELINE=(0.02,2.0,2.0)) -> harness.py (DB-pred scorer) -> batch.py (corpus aggregate) -> regression.py (CI gate). calibration.py = the overfit guard (cross_validate, leave_one_video_out, generalization_gap). Learned segmenter: features/rally_seq.py (the single train==serve feature schema, ADR-008) -> rally_seq_proto.py (Gen-0 LOVO prototype) / training.py + classifier.py (numpy logreg). A/B layer = experiment.py.\n\nOther subsystems: stitcher/compiler.py (re-encode-then-concat, optional branding watermark), validators/ (HSV relevance, blur, scene-cut, PTS), sports/ + providers/ + annotations/ registries, infrastructure/database.py (6 tables), reporting/ (per-video obsreport: harvest.py records facts, spec.yaml = declarative footgun rules coupled by dotted path to harvest's stage/metric/detail names). Section 7 RAMP PATHS maps \"to do X -> open Y::symbol\"; Section 8 TEST MAP maps each tests/test_*.py to its subsystem. Repo-placement rules in Section 0; data contracts (trajectory CSV, candidate window dict, play_segment, jobs row, GT CSVs, eval primitives) in Section 3; WASB WSL contract in Section 4; cross-repo (sports-data-collector keys by human video_id, indexer by file MD5) in Section 5; cross-cutting footguns in Section 6.

"served_operating_point": "\"Served\" = the operating point/config the live (production) pipeline actually runs with - the A side of every A/B comparison. Concretely it is the DEFAULT config with all the experimental \"milestone\"/fusion knobs OFF, i.e. the pinned baseline that current users get.\n\nIt surfaces in three coupled places in CODE_MAP.md:\n\n1) Nightly rally-quality tripwire (scripts/nightly_regression.py): re-scores the golden corpus under TWO configs and diffs them - \"default (SERVED, milestone knobs OFF)\" vs \"milestone (R3 cap=6 + R4 inactivity trim)\". So the SERVED preset is explicitly the default config without the milestone over-seg-fighting knobs. (A future RAW eval set plugs in as a third config.) This is default-ADDITIVE and does NOT change the promotion gate.\n\n2) The A/B / experiment layer (backend/eval/experiment.py): ::CONTROL is \"the pinned zero-gate baseline\" - the A side. A Variant is the B side (segmenter + config_overrides + cue_flags + Gate A min_crossings + Gate B min_players + optional learned classifier/threshold). ::compare reports labeled B-A deltas; ::compare_unlabeled does side-by-side on new footage with no GT. The tested no-regression invariant is: ALL CUES OFF => byte-identical to CONTROL. So the served operating point IS CONTROL = all gates/cues at zero.\n\n3) The fusion segmenter (backend/pipeline/segmenters/fusion_hybrid.py): ships default-OFF. Its Gate A (fusion.min_crossings) and Gate B (fusion.min_players, person-only) default to 0/0, so all windows are kept and behavior is byte-identical to the served wasb path. Persisted candidates are ALWAYS the full superset; only the handoff/play_segments are gated. To A/B a fusion run fairly against a tuned WASB run you must set fusion.velocity_threshold = wasb.velocity_threshold (the engine reads idx_cfg['fusion'].velocity_threshold).\n\nThe promotion discipline this enables: new cues/gates/learned segmenters land flag-gated and default-OFF so the SERVED point is unchanged and regression-free; the experiment harness measures the candidate (B) against the served control (A) - on labeled golden via B-A IoU/F1 deltas, and on unlabeled new footage via agreed/a_only/b_only SxS - before anything is promoted into what is served. The segmentation knobs being tuned at this operating point are the three from the mental model (velocity_thresh, merge_gap, min_window_duration), with BASELINE=(0.02,2.0,2.0) in calibrate_wasb as the reference served threshold triple.

"files_found": [

```

"/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/HOW_RALLY_DETECTION_WORKS.md",
"/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/CODE_MAP.md"
],
"code_map": "Both requested files EXIST and were read in full. Neither is missing.\n\nKey
navigation anchors (all paths relative to /home/avidullu/projects/khelsutra-guru/badminton-
highlight-indexer/):\n- HOW_RALLY_DETECTION_WORKS.md: the 2-layer mental model (frozen WASB
DETECTION (x,y) -> velocity -> 3-knob SEGMENTATION -> [start,end] -> temporal IoU vs golden).
Golden labels can only tune the segmentation layer.\n- CODE_MAP.md: cold-start nav map.
Sections: 0 REPO LAYOUT (where new files go), 1 PIPELINE (1A runtime, 1B WASB WSL substrate, 1C
calibration/eval), 2 SUBSYSTEMS (2.0 orchestration/config ... 2.8 reporting), 3 DATA CONTRACTS,
4 WASB WSL CONTRACT, 5 CROSS-REPO, 6 GOTCHAS, 7 RAMP PATHS (task->symbol), 8 TEST MAP.\n\nLoad-
bearing symbols cited:\n- Windowing brain (the 3 knobs run here):
backend/pipeline/detectors/tracknet_runner.py::trajectory_to_action_windows (staticmethod, re-
exported by WasbRunner).\n- Live shuttle substrate:
backend/pipeline/segmenters/wasb_hybrid.py::WasbHybridSegmenter (MIT, pretrained badminton
weights); WSL GPU inference backend/pipeline/detectors/wasb_infer.py.\n- Served baseline / A
side: backend/eval/experiment.py::CONTROL (pinned zero-gate); compare / compare_unlabeled;
invariant \"all cues OFF => byte-identical to CONTROL\".\n- Served vs milestone diff:
scripts/nightly_regression.py (default=SERVED, milestone-knobs-OFF vs milestone config).\n-
Default-OFF fusion: backend/pipeline/segmenters/fusion_hybrid.py::FusionSegmenter (Gate A
fusion.min_crossings, Gate B fusion.min_players, both default 0 => served-identical).\n-
Reference threshold triple: backend/eval/calibrate_wasb.py::BASELINE=(0.02,2.0,2.0).\n- Eval
spine: backend/eval/metrics.py; overfit guard backend/eval/calibration.py (leave_one_video_out,
generalization_gap); CI gate backend/eval/regression.py + run_regression.py (exit 0 ok / 1
regress / 2 no-DB / 3 no-baseline).\"
}
},
{
"key": "doc-protocol-quality",
"data": {
"doc_status_protocol": "DOC_STATUS.md archival due-diligence protocol (the honest
update->archive checklist, section 2). Run BEFORE any foundational/tracked doc moves to
docs/archives/ (do NOT just git mv): 1) VERIFY claims against the code -- no stale assertions;
tag load-bearing facts [verified]. A doc that says \"nothing built\" when code shipped is NOT
archive-ready -- fix it first. 2) FLIP Status -> DELIVERED / DONE / REJECTED + a completion note
-- what shipped, where (PRs, paths, artifacts), and what was deferred / carried forward. 3)
UPDATE ALL inbound references -- docs/README.md index, CLAUDE.md, docs/NEXT_STEPS.md, and any
cross-doc links -> new path + status. 4) KEEP THE LINEAGE -- ADR forward/back-links +
\"supersedes / superseded-by\" so the evolution stays traceable. 5) git mv the whole dir/file ->
docs/archives/past_projects/ (terminal-state ONLY; live docs stay at docs/ top level). Relative
links inside become historical (note this in the doc header). 6) RECORD THE MOVE -- flip the row
to register section 3e + a line in memory current-state. Worked template: DECOUPLED_COMPUTE ->
#270 (verified M0-M2+M3.5, flipped to DELIVERED, reindexed READMEs, kept ADR-010->014 lineage,
then git mv). Lifecycle legend (section 1): DRAFT -> IN PROGRESS -> DONE/DELIVERED -> ARCHIVED;
plus LIVING, PROPOSAL/DESIGN, REPORT, CONVENTION. Update cadence: whenever a doc changes
lifecycle state; re-review section 4 at each clean-slate. Owner directive (2026-06-21): never
archive a foundational doc without honestly updating it first. DOC_STATUS.md itself is status
LIVING, last full review 2026-06-21.</doc_status_protocol>\n\n<parameter
name=\"quality_table\">RALLY_DETECTION_QUALITY_REPORT.md -- checkpoint 2026-06-17, golden corpus
n=11 (5 with real ground truth), supersedes 2026-06-14 draft. OVER-SEGMENTATION MILESTONE
(\"cap=6 + R4\", n=11, R2 task-faithful tolerance-F1, tau_start=2.0s / tau_end=1.5s):
tolerance-F1 0.091 -> 0.323 (delta +0.232, 95pct CI [+0.153,+0.304], paired sign-test 10/0/1,
p=0.002) -- first statistically significant rally-quality win, served-measured. Merge-rate
(golden rallies swallowed by a merge) 0.694 -> 0.150 (down on all 11 videos, p=0.001). Mean
rally-END error 52.1s -> 3.8s. Windowing ladder (n=11, tolerance-F1 [95pct CI] | tIoU-F1
secondary | mean |dEnd|): baseline all-OFF 0.091 [0.026,0.179] | 0.195 | 52.1s; +w1 cap=12 0.168
[0.100,0.241] | 0.357 | 6.5s; +cap=6 (R3) 0.276 [0.174,0.380] | 0.407 | 5.7s; +R4 milestone
0.323 [0.210,0.433] | 0.440 | 3.8s. PER-VENUE REFERENCE (rally-detection F1 @ IoU 0.5 vs human
golden labels): Heuristic velocity windowing -- Mahadevpura(clean) 0.657, Kushagra(multi-court)
0.157; Gen-0 owned learned head (LOVO n=6) -- 0.744 / 0.561; Two-stage WASB+Gemini refine --
0.83 / not measured; Gemini wrapper (gemini-flash-latest) -- 0.93 / 0.66. REAL-FOOTAGE GROUND
TRUTH (n=2): local ~86-88pct of Gemini on detection (F1@0.25); rally-START at Gemini parity;
entire residual gap is rally-END (local |dEnd| 2.1-3.1s vs Gemini ~0.8-1.2s). FUSION ABLATION
(n=6, LOVO, IoU 0.5): shuttle-only 0.307; +person 0.286 (dF1 -0.021, p=0.688, no); +person+audio

```

```

0.366 (dF1 +0.079, p=0.688, no) -- both stay default-OFF, not promotable at
n=6.</quality_table>\n<parameter name=\"ship_target\">SHIP TARGET = 0.66 -- the Gemini cloud
wrapper's multi-court (Kushagra) F1 @ IoU 0.5. \"Clean footage is commoditized (Gemini serve =
0.93); the multi-court 0.66 is the line the owned model must reach to be economically
justified.\" Gen-0 owned head is at 0.561 multi-court (n=6) -- 0.10 / ~0.099 behind, described
as a corpus-growth-sized gap, not an architecture-sized one. Heuristic LOVO floor across n=6 =
0.480. Gen-0 aggregate LOVO 0.626 (sign-test 5/6, p=0.109 -> not yet significant, ship=False).
NOTE: the formal ship-gate target number is still owner-gated / unset (NEXT_STEPS P1 item #7
\"Set the ship-gate target #172\" + P0 #172 -- both need corpus growth). NEXT_STEPS suggested
shape (owner to ratify): \"LOVO F1@tIoU0.5 >= 0.70 on multi-court folds with paired-sign p<0.05
vs heuristic\".</ship_target>\n<parameter name=\"next_steps_open_items\">OPEN / SHIP-BLOCKING
(from NEXT_STEPS.md refreshed 2026-06-21 + report section 7). P0 highest-leverage: (#0) owner-
priority -- make it UI-driven + alpha-shareable (recordings -> highlights from a page, no SSH);
needs M7 Cloud Run service [owner GCP spend], khelsutra web/ SPA ingest screen, flip M9 edge-
auth ON, compute PICKER in UI. (#0b) packaging track IN PROGRESS (11 PRs merged) -- downloadable
batteries-included app; next P3 GPU validation+screencast (WARNING native-WASB 0-rally tuned-
config blocker), P2b-install, P2c first-run wizard. (#1) COMPUTE_DECOUPLED_SERVING M0-M9 largely
shipped; remaining M11 + M12 (P1 #4). (#2) [owner, ongoing] GROW THE GOLDEN CORPUS -- multi-
court badminton + >=3 matches/sport (table tennis first); THE single highest-leverage move, the
#1 lever, everything feeds on it. (#3) [owner+Claude GPU] Full-corpus training n>=5 -> real
weights/ bundle. P1 near-term: (#5) per-video workspace P1-P6; (#7) OWNER -- SET THE SHIP-GATE
TARGET (#172), needs corpus growth; (#8) [owner GPU] R-series promote-or-reject R2 toLF1 as the
gate (the ablation), needs GPU-extracted golden features. P2 later/gated: (#9) rally-quality
P3/P4 person-fusion LOVO + audio cue (needs GPU golden features); (#9b) [owner GPU golden]
promote the R4 density fix (PR #393 landed default-OFF behind inactivity_temporal_density; ship
= golden A/B ON vs OFF at SERVED windowing, flip to True if toLF1 doesn't regress); (#10) M0.4
scorer swap TemporalMaxer; (#11) multisport TT then pickleball; (#13) WASB-C3 weight provenance
(gates sharing a trained model; legal); (#14) rename repo off \"badminton\". REPORT section 7
planned wins / levers: grow corpus (n>=16, multi-court isolated as own held-out slice);
milestone cap=6+R4 = DONE/LAUNCHED PR #204 2026-06-18 (shipped default max_window_duration=6.0,
window_inactivity_end=1.5); close rally-END gap (|dEnd| 3.8s -> Gemini ~1s via stronger
sustained-inactivity END rule); activate court polygons (fixes person-cue spectator noise); re-
measure person/audio on bounded windowing; ActionFormer/TemporalMaxer swap (gated on corpus
growth); owned VLM (Qwen3-VL Apache-2.0 / InternVL3 MIT fallback). COMMERCIAL-SHIP BLOCKER: C3
WASB checkpoint provenance -- an owned or cleanly-licensed detector is required before
commercial release; a clean head does not launder it. DOC-HYGIENE pending (DOC_STATUS section
4): confirm COMPLETED top-level stubs (CODE_AUDIT_AND_TEST_HARDENING / HARDENING_LOOP_HANDOFF /
DEFERRED_HARDENING_PLAN) are intentional pointer-stubs vs finish-archive; add HOSTING_PLAN.md
currency banner (still pending; partly overtaken by ADR-013 + shipped site); R-series/quality
cluster ACTIVE -- do NOT archive yet (archive as a cluster once R2 promoted-or-rejected as the
gate + section 7.1 single quality number is attributed).</next_steps_open_items>\n<parameter
name=\"files_found\">[\"/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/DOC_STATUS.md (EXISTS, 11972 bytes)\", \"/home/avidullu/projects/khelsutra-
guru/badminton-highlight-indexer/docs/RALLY_DETECTION_QUALITY_REPORT.md (EXISTS, 32393 bytes)\",
\"/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/NEXT_STEPS.md (EXISTS,
22619 bytes, refreshed 2026-06-29 per mtime / header says refreshed 2026-06-21)\"]\"
}
},
{
  \"key\": \"test-infra-guardrails\",
  \"data\": {
    \"run_all_tests\": \"{\\n  \\\"file\\\": \\\"/home/avidullu/projects/khelsutra-guru/badminton-
highlight-indexer/run_all_tests.py\\\",\\n  \\\"what_it_runs\\\": \\\"Single discoverable entry point
that shells out to `python -m pytest` (subprocess.run([sys.executable, '-m', 'pytest', *args])).
It passes ALL CLI args after the script name straight through to pytest. Defaults to quiet (-q)
unless caller passes -q/-v/-vv/-vvv/--quiet/--verbose. Exit code is pytest's own (0 = all
green), so it is CI/pre-PR friendly.\\\",\\n  \\\"scope\\\": \\\"pytest.ini sets `testpaths = tests`, so
a bare invocation collects only from tests/ (scratch/ holds real-API debugging scripts and is
never collected). Suite is fully mocked: no GPU, no network, no WSL, runs offline.\\\",\\n
\\\"obsreport_probe\\\": \\\"Before running, it probes importlib.util.find_spec('obsreport') and
prints whether the optional reporting dep is available or 'NOT installed -> reporting tests will
SKIP'. It does NOT install it; reporting tests self-skip via
pytest.importorskip('obsreport').\\\",\\n  \\\"typical_invocations\\\": [\\n    \\\"python
run_all_tests.py # full suite, quiet\\\",\\n    \\\"python run_all_tests.py -v
# verbose passthrough\\\",\\n    \\\"python run_all_tests.py -k config # keyword filter

```

```

passthrough",\n    \python run_all_tests.py --cov=backend --cov-report=term-missing #
coverage"\n ],\n    \ci_invocation": \CI/nightly call it as: python run_all_tests.py
--cov=backend --cov=training --cov-report=term --cov-fail-under=70"\n}",
"gpu_realvideo_golden_gating": "{\n    \summary": \There are NO registered pytest markers
(pytest.ini has only \testpaths`; no [pytest] markers/addopts in pytest.ini, pyproject.toml, or
setup.cfg) and there is NO conftest.py anywhere in the repo. GPU / real-video / golden-corpus /
native-WASB gating is done entirely via (a) pytest.importorskip for optional deps, (b)
pytest.mark.skipif on data-presence predicates, and (c) monkeypatch.setenv to drive code paths
through CPU-only stubs. The whole suite is designed to run offline with no GPU.\",\n
\gpu_gating\": {\n    \mechanism\": \No GPU is ever required by the offline suite. GPU-
touching code is exercised through mocked subprocesses / CPU stubs. Real GPU byte-parity is
explicitly handed to the owner box, not run in pytest.\",\n    \examples\": [\n
\"tests/test_native_wasb_runner.py: native subprocess mocked (no torch/conda/GPU); device=auto
on a GPU-less box resolves to cpu and warns (does not hard-fail); the lazy cuda probe is mocked
(empty stdout = no cuda).\",\n    \tests/test_worker.py:
monkeypatch.setenv('RALLY_DETECTOR_IMPL','stub') selects a CPU stub runner so wiring is
exercised GPU-free; 'native' on a family with no native runner returns clean rc 2.\",\n
\"tests/test_fusion_golden_telemetry.py: stubs extract_video (no GPU), monkeypatches rt.gpu_name
-> 'FakeGPU' (no real nvidia-smi).\",\n    \tests/test_fusion_features.py,
tests/test_serve_contrast.py: explicitly 'No GPU/torch/cv2' ? synthetic person boxes + shuttle
points exercise the fusion math.\",\n    ]\n },\n    \real_video_golden_gating\": {\n
\mechanism\": \pytest.mark.skipif on a predicate that checks whether the gitignored golden
corpus / manifest / feature JSONs are physically present. Absent in CI / a fresh worktree -> the
test SKIPS cleanly; present on the owner box -> it runs.\",\n    \examples\": [\n
\"tests/test_served_gate_a_regression.py: builds _SPECS via _golden_specs() reading the manifest
at env SERVED_GATE_A_MANIFEST (default <repo>/output/human_lovo_manifest.json); requires >= 6
specs whose trajectory+labels files all exist, else []. _skip = pytest.mark.skipif(not _SPECS,
reason='golden trajectories absent ... runs on the owner box')`. Module also gated by
pytest.importorskip('cv2').\",\n    \tests/test_ablation.py: pytest.mark.skipif(not
_golden_present()) where _golden_present() counts output/*_golden_features.json with the fusion
schema and requires n >= 6; the LITMUS test reproduces hand-measured Gate-A (delta_f1 ~ +0.074,
6/6 wins, p ~ 0.031) only when those 6 real golden feature JSONs exist.\",\n
\"tests/test_golden_fixtures.py + tests/test_golden_real_fixtures.py: COMMITTED, de-attributed
synthetic+real trajectory fixtures (no video/GPU/DB). These DO run in CI (the golden-crossos /
golden-fixtures lanes) via \python -m backend.eval.golden_fixtures --check\".\",\n    ]\n },\n
\native_wasb_gating\": {\n    \file\": \tests/test_native_wasb_runner.py\",,\n
\mechanism\": \NativeWasbRunner is the Linux-native (no-WSL) WASB detector path (M3.5). Tests
run with the native subprocess fully MOCKED ? no torch/conda/GPU. They lock the contract that
the GPU-box byte-parity gate later relies on: it IS a DetectorRunner, carries NO WSL plumbing
(no WslCommandMixin / no \wsl` / no \conda activate` in argv), builds a native argv, honors
WASB_* env overrides (WASB_WEIGHTS/WASB_DEVICE/WASB_PYTHON/WASB_REPO_DIR/WASB_SCRATCH) and
config; default device is STREAMING. Selection seam driven by config \detector_impl='native`
and env RALLY_DETECTOR_IMPL='native' (refused for tracknet / fusion-tracknet families). The
actual byte-for-byte GPU parity diff is NOT a pytest test ? it is the owner-box gate; pytest
only stages/asserts the wasb_infer.py is byte-identical to the WSL runner's
(test_copy_infer_script_copies_byte_identical).\",\n    \tracknet_note\":
\"tests/test_person_detector.py gates the D13/P12 ByteTrack tests on
pytest.importorskip('trackers') (git-only dep).\",\n    ],\n    \importorskip_gated_deps\": {\n
\httpx\": \FastAPI TestClient transport ? tests/test_api_endpoints.py, test_async_jobs.py,
test_upload_download.py, test_bind_and_host.py, test_edge_auth.py, test_ui_bundle.py,
test_job_worker_loop.py, test_api_cloud_dispatch.py\",,\n    \obsreport\": \Optional private
reporting dep ? tests/test_reporting_harvest.py, test_reporting_spec_faq.py (run_all_tests
prints availability)\",,\n    \cv2\": \tests/test_wasb_cache.py,
test_served_gate_a_regression.py, test_nightly_reelcount.py, test_rally_seg_eval.py\",,\n
\trackers\": \tests/test_person_detector.py (D13/P12 ByteTrack)\",,\n    \reportlab + pypdf\":
\"tests/test_ablation.py PDF report tests\",,\n    \PIL.Image\": \tests/test_wasb_cache.py\",,\n
\numpy + scipy\": \tests/test_import_smoke.py\",,\n    \PyJWT/jwt\": \tests/test_edge_auth.py
(M9 edge-auth, lazy-imported)\",,\n    ],\n    \env_var_drivers\": [\n    \RALLY_DETECTOR_IMPL
(stub|native) ? selects CPU stub vs native runner without GPU/WSL\",,\n    \WASB_WEIGHTS /
WASB_DEVICE / WASB_PYTHON / WASB_REPO_DIR / WASB_SCRATCH ? native runner config overrides\",,\n
\SERVED_GATE_A_MANIFEST ? path to the golden manifest for the served-gate-A regression\",,\n
],\n    \tiering\": \Per nightly.yml header: Tier-1 = OFFLINE ONLY (no GPU, no real video, no
run_regression.py on the real corpus). Tier-2 (real video + full GPU/WSL/WASB byte-parity suite,
golden corpus + labelled videos) is owner-gated on a self-hosted GPU runner ? handed to the
owner, per CLAUDE.md.\",\n    ]\n }",

```

```

"claude_md_guardrails": "{\n  \"file\": \"/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/CLAUDE.md\",\n  \"commercial_clean_licensing\": [\n    \"Commercial-clean licensing for EVERY dependency ? code, data, AND weights: MIT / Apache-2.0 / BSD ONLY (ratified 2026-06-09, owner-agreed).\",\n    \"Owned-model base: Qwen3-VL (Apache-2.0) primary, InternVL3 (MIT) fallback. Gemma-4 is AMBER / bench-only (Apache grant likely but Prohibited-Use posture unconfirmed; counsel needed; do NOT build on it without sign-off). REJECT Gemma 3 (viral license).\",\n    \"Enforced in CI by license-scan lane (ci.yml + nightly.yml): scans the runtime dependency closure with pip-licenses and FAILS on any copyleft (grep -iE 'GPL|SSPL').\",\n    ],\n  \"paid_llm_inference_only\": [\n    \"Paid LLMs (Gemini / OpenAI / Anthropic) = inference / serving / fallback ONLY ? NEVER a training data source (terms/copyright).\",\n    \"Exclude minors from the training pool; per-user training data needs a separate explicit opt-in.\",\n    \"Paid Gemini is the cloud AI for serving/inference; the owned-model track must train on commercial-clean owned data, not paid-LLM outputs.\"\n  ],\n  \"owner_gated_work\": [\n    \"No platform / owned-model implementation has started; do NOT start implementation without explicit owner approval.\",\n    \"Two tracks, both owner-gated: committed next PLATFORM slice = async job model (DEFERRED_HARDENING_PLAN.md #16, single-tenant); owned-model track starts at a greenlit Phase-0 milestone.\",\n    \"Rally-detection quality track is owner-gated (design merged 2026-06-13, implementation in progress).\",\n    \"Hand the heaviest machine-side checks (real video, the full GPU/WSL/WASB suite) to the owner, who has the golden corpus + labelled videos; verify pure-logic paths locally.\",\n    \"Substantial design/project work gets a tracked project doc following docs/PROJECT_DOC_TEMPLATE.md.\"\n  ],\n  \"branch_pr_rules\": [\n    \"Branch off master.\",\n    \"ONE PR per work item; NO PR stacking.\",\n    \"Do NOT open a PR unless asked.\",\n    \"Never push to the default branch directly; use descriptive commit messages.\",\n    \"Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.10+ (the trackers package requires >=3.10); ffmpeg required.\"\n  ],\n  \"scope_guardrails\": [\n    \"v1 = rally + highlight + history + correction loop; gait/biometrics strictly FUTURE (GDPR Art. 9).\",\n    \"Generic data schema; coach -> athlete is an OPTIONAL overlay, not baked in.\",\n    \"Compute = GCP ONLY (ADR-013, 2026-06-20): GPU training + serving on GCP (Cloud Run+GPU for serving per ADR-012). DigitalOcean DROPPED for compute; Paperspace subscription-gated. Don't re-evaluate DO/Paperspace without a new ADR; $200 DO credits go to non-compute only. deploy/digitalocean/* kept only as provider-agnostic Linux-GPU setup reused by GCP.\"\n  ],\n  \"archive_policy\": \"Only move terminal-state (DONE/REJECTED) work to docs/archives/; live docs stay at docs/ top level. Before archiving, HONESTLY update the doc first (verify claims vs code -> flip status + completion note -> update inbound refs -> keep lineage -> git mv), per docs/DOC_STATUS.md.\"\n}\n",
"ruff_mypy_config": "{\n  \"ruff\": {\n    \"config_file\": \"/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/ruff.toml\",\n    \"ci_invocation\": \"ruff check . --output-format github (CI lane 'lint' in ci.yml + 'Nightly · Ruff lint' in nightly.yml; runs on Python 3.13 runner with `pip install ruff`)\",\n    \"mode\": \"Lint-only ? NO `ruff format` enforcement (a whole-repo reformat would bury blame).\",\n    \"target_version\": \"py38\",\n    \"selected_rules\": \"ruff defaults (pyflakes F + pycodestyle errors E) PLUS extend-select = [W (pycodestyle warnings), B (flake8-bugbear), I (isort import ordering)].\",\n    \"excludes\": \"extend-exclude = [scratch] (throwaway debugging scripts; also excluded from pytest collection).\",\n    \"per_file_ignores\": \"E402 (module-import-not-at-top) for backend/main.py, scripts/run_phase3_eval.py, scripts/run_process_and_stitch.py (load-bearing setup-before-import order).\",\n    \"also_serves_as\": \"repo-wide syntax guard ? ruff reports invalid-syntax on every tracked .py incl. scripts/ and training/.\"\n  },\n  \"mypy\": {\n    \"config_file\": \"/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/mypy.ini\",\n    \"ci_invocation\": \"mypy backend training (CI lane 'typecheck'/'Nightly · Mypy (lenient baseline)'); Python 3.13 runner with `pip install mypy pydantic fastapi uvicorn python-dotenv google-genai numpy` so real signatures are seen).\",\n    \"baseline\": \"Lenient, GREEN baseline (quality sweep AI-4) ? protects clean modules with an explicit per-module quarantine; ratchet by removing entries when you touch a quarantined module.\",\n    \"global_settings\": {\n      \"ignore_missing_imports\": True; explicit_package_bases = True (backend is a namespace package with no root __init__.py).\n    },\n    \"quarantined_modules_ignore_errors\": [\n      \"backend.main\",\n      \"backend.worker.__main__\",\n      \"backend.pipeline.segmenters.yolo_hybrid\",\n      \"backend.pipeline.segmenters.gemini\",\n      \"backend.pipeline.segmenters.base\"\n    ]\n  },\n  \"coverage_floor\": {\n    \"floor\": \"--cov-fail-under=70 (70%)\",\n    \"enforced_where\": \"ci.yml::full-suite and nightly.yml::full-suite, identical literal command: `python run_all_tests.py --cov=backend --cov=training --cov-report=term --cov-fail-under=70`\",\n    \"calibration_note\": \"Per ci.yml comments: floor calibrated 2026-06-11 against the CI lane's measured 72% (obsreport ABSENT, reporting tests skip); local with obsreport installed measures ~73%. Ratchet UP as coverage grows; NEVER lower without an owner decision. The literal is copied

```

```
(not single-sourced) between ci.yml and nightly.yml ? a noted follow-up.\",\n
\"coverage_config_file\": \"~/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/.coveragerc\", \n \"coverage_omits\": [\n \"backend/pipeline/segmenters/motion.py
(legacy demo baseline with fabricated metadata)\", \n
\"backend/pipeline/validators/format.py\", \n
\"backend/pipeline/validators/resolution_fps.py\", \n
\"backend/pipeline/validators/pts_continuity.py\", \n
\"backend/pipeline/validators/scene_cut.py\", \n \"backend/pipeline/validators/blur.py\", \n
\"backend/pipeline/validators/relevance.py\" \n ], \n \"coverage_omit_note\": \"Concrete
validators are thin OpenCV/ffprobe wrappers only exercisable against real media;
validators/base.py stays counted (it has a dedicated unit suite).\\n\\n}\",
\"expected_test_count\": \"{ \n \"documented_latest\": \"1492/1492 green
(docs/CODE_CLEANUP_AUDIT_2026-06-24.md, dated 2026-06-25, P9/B1 entry). Preceding documented
checkpoints in the same audit log: 1463/1463, 1484/1484. The prompt's '~1487+' aligns with this
mid-1480s-to-1490s range.\", \n \"documented_progression\": [\"Suite 1068\", \"Suite 1105\",
\"Suite 1114\", \"Suite 1126\", \"1365 pass\", \"1390 pass\", \"Suite 1463 / 1463/1463 green\",
\"Suite 1484 / 1484/1484 green\", \"Suite 1492 / 1492/1492 green (latest)\"] \n
\"actual_collected_local\": \"1513 tests collected (this checkout, via `python -m pytest
--collect-only -q`) ? higher than the documented 1492 because this local environment has the
optional deps present (cv2/httpx/obsreport/trackers/numpy/scipy etc.), so importorskip-gated and
parametrized tests are NOT skipped/are fully expanded. Raw `def test_` defs = 1390; the gap to
1513 is pytest.mark.parametrize expansion. No async tests (0).\", \n \"ci_lane_count_caveat\":
\"The CI full-suite lane number is lower than a fully-equipped local run because obsreport
(private) is absent in CI -> reporting tests skip via pytest.importorskip, and golden/owner-box
tests skip on absent corpus. The 70% coverage floor is calibrated against the CI lane number,
not the local one.\", \n \"where_documented\": \"Not stated in CLAUDE.md, README.md, or
run_all_tests.py as a fixed target; the running suite count is recorded as a green-checkpoint
trail in docs/CODE_CLEANUP_AUDIT_2026-06-24.md and other docs (e.g. docs change-logs).\\n\\n}\"
}
}
]
```

Produce a tight Markdown runbook with these sections:

1. CURRENT STATE ? P1/P2a/P2b status from the §7 tracker; current default of inactivity_temporal_density and its exact file:line.
2. FINDING B ? one-paragraph plain explanation + why the fix is a no-op under dense tracking.
3. A/B RUNBOOK ? the exact commands in order (fusion_golden feature-gen, then rally_seg_eval served vs served with the temporal-density flag), with placeholders for the manifest/features paths clearly marked.
4. DECISION RULE ? headline metric (tol_f1), the promote-vs-reject rule, served_regress_eps, secondary metrics to inspect, and the promote_if_served_safe gate.
5. ON PROMOTE ? exact one-line code change + test update + doc updates (§7 ? DONE, DOC_STATUS archive protocol).
6. ON REJECT ? leave flag OFF, record numbers, §7 ? REJECTED.
7. FULL VALIDATION ? run_all_tests.py, the now-unlockable GPU/real-video/golden/WASB test gates, ruff + mypy + coverage floor.
8. GUARDRAILS ? CLAUDE.md hard rules + branch/PR discipline.
9. OPEN QUESTIONS / UNKNOWNNS ? anything the digests could not resolve (missing files, unknown paths).

Be precise and command-accurate. This is my working reference, output the Markdown only.

2. assistant (2026-06-29T10:54:00.162Z)

This is a synthesis task ? I have all the structured digests I need to assemble the runbook. No tool calls required; I'll produce the Markdown directly.

P2b Runbook ? Promote `inactivity\_temporal\_density` (R4 finding-B fix)

```
> **Source of truth:** ~/home/avidullu/projects/khelsutra-guru/badminton-highlight-
indexer/docs/RALLY_DETECTION_FIXES_PLAN.md` §7 tracker.
> **Extends:** `docs/RALLY_DETECTION_QUALITY_REPORT.md`. **Code-review backing:**
`docs/RALLY_DETECTION_CODE_REVIEW.md` (finding B).
> All paths below are relative to ~/home/avidullu/projects/khelsutra-guru/badminton-highlight-
```

indexer/` unless absolute.

1. CURRENT STATE

From the S7 tracker ("Deliverables & progress tracker ? source of truth"; legend ? Todo . ? In progress . ? Done . ? Blocked/gated):

Row	Item	Dep	Gated?	Status	PR
***P1**	Warn on rows silently skipped by `load_canonical_csv` (sibling `sports-data-collector`)	?	No	? Done	`sdc#50`
***P2a**	Expose the flag in the `rally_seg_eval` A/B CLI (`--[vs-]inactivity-temporal-density`)	P0b	No	? Done	`this PR`
***P2b**	***Promote P0b**:	golden-set A/B at SERVED windowing ? flip flag default-ON if no regression	P2a	? owner	? **Todo** `?` (none)

- **P2b is the only open/blocked row.** Its dependency P2a is done; promotion is **owner-gated** because it needs the golden corpus + GPU/WSL-extracted trajectory features that do not exist offline, and finding B only shows signal on clips where the shuttle tracker drops out (not measurable on synthetic/offline fixtures).

- **Current default of the flag:**

...

backend/config/models.py:199

... inactivity_temporal_density: bool = False

...

Class `IndexingConfig(BaseModel)` (declared `backend/config/models.py:146`). Default **`False` (OFF)**. The runner param mirrors this at `backend/pipeline/detectors/tracknet\_runner.py:329` (`inactivity\_temporal\_density: bool = False`).

> **Doc inconsistencies to be aware of (do not "fix" silently):** (a) S7 lists P0a/P0b under PR **394** while the S0/header narrative cites **393**;

(b) S7 marks P1 ? Done (`sdc#50`) but the S9 Definition-of-Done `rally\_cvat.py` warning-log checkbox is still unchecked.

2. FINDING B

Plain explanation. R4 ("Inactive Tail Trimming", `backend/pipeline/detectors/tracknet\_runner.py::\_trim\_inactive\_tail`, def line 450) trims a candidate rally window's **end** back to the last frame whose trailing `trail\_frames` window holds enough "fast" frames (velocity > `rally\_thresh`). The legacy density check is `fast\_sum / win\_count >= min\_density`, where `win\_count` is the count of **visible/active samples** in the trailing window ? **not** elapsed time. If the detector loses the shuttle for ~1.8 s and then catches a single fast frame at the very end, `win\_count` is tiny (172) so `fast\_sum/win\_count` reads 1.0 (100% dense). Sparse trailing points then **defeat the trimmer**, leaving dead-time uncut and artificially inflating the rally **end** boundary. The fix adds the `inactivity\_temporal\_density` flag: when **ON**, the density **denominator** becomes the trailing window's **temporal length** `min(trail\_frames, frames[i] - frames[0] + 1)` instead of the active-sample count, so sparse tracking can no longer inflate density. Resolved as strict temporal density (no separate few-frames fallback). R4 itself is **default-ON in prod** (`window\_inactivity\_end = 1.5`), which is why this real behaviour change ships behind a default-OFF flag.

Why the fix is a no-op under dense tracking. The `group` passed to `\_trim\_inactive\_tail` contains **only active frames**. Under dense tracking every frame in the trailing span is active, so the temporal denominator `min(trail\_frames, frames[i]-frames[0]+1)` **equals** the sample-count denominator `win\_count`. Identical denominators ? identical output ? flipping the flag changes nothing on well-tracked clips. Behaviour diverges **only** in the sparse-tracking case. Corollary: signal appears only where the shuttle tracker actually drops out, so the A/B cannot be measured on synthetic/offline fixtures ? hence the owner gate on real labelled video.

3. A/B RUNBOOK

```
> **Owner box only** (real labelled video + GPU/WSL). The whole point is to compare base =  
SERVED **flag-OFF** vs candidate = SERVED **flag-ON** at the SERVED operating point.  
> **Placeholders ? replace before running:**  
> - `<GOLDEN_MANIFEST>` = path to the golden manifest JSON (no concrete path in the plan doc;  
the served-gate-A regression default is `<repo>/output/human_lovo_manifest.json` via env  
`SERVED_GATE_A_MANIFEST`).  
> - `<GOLDEN_FEATURES_DIR>` = directory of GPU/WSL-extracted golden trajectory features (no  
concrete path in the plan doc; fusion golden features land as `output/*_golden_features.json`).
```

****Step 1 ? Generate golden trajectory/fusion features (GPU/WSL).**** These are gitignored and must be extracted on the owner box before the A/B can see signal.

```
```bash
```

**Extract the golden features the A/B reads from (GPU/WSL; produces output/\*\_golden\_**  
**Adjust the manifest/output paths to your corpus layout.**

```
python -m backend.eval.golden_fixtures --check # sanity: confirms fixture schema is
intact
```

**(feature-gen is the owner-box GPU/WSL extraction step; ensure `<GOLDEN_FEATURES_DIR>`**  
**is populated with the per-video trajectory feature JSONs before Step 2)**

```
...
```

```
> The plan doc does not pin an exact feature-gen command; the offline `golden_fixtures --check`
is the committed sanity gate. The actual trajectory-feature extraction is the WSL-quarantined
WASB inference path (`backend/pipeline/detectors/wasb_infer.py`), which is **not importable on
Windows** and runs in WSL2 conda.
```

**\*\*Step 2 ? Run the promotion A/B (the canonical P2b command, §8 Q2).\*\***

```
```bash
```

```
python -m backend.eval.rally_seg_eval \  
    --manifest <GOLDEN_MANIFEST> --features-dir <GOLDEN_FEATURES_DIR> \  
    --preset served --vs served --vs-inactivity-temporal-density  
...
```

- Both sides resolve to the ****served**** preset (`--preset served --vs served`); only the `--vs`
(candidate) side flips `inactivity\_temporal\_density=True`, isolating the finding-B denominator
change at the served operating point.

- Wiring: `main()` builds `base = \_build\_preset(args, "base")` and `cand =
_build_preset(args, "cand")`. For the candidate, `temporal_density =
args.vs_inactivity_temporal_density` or `args.inactivity_temporal_density` then `replace(preset,
inactivity_temporal_density=True)` (`backend/eval/rally_seg_eval.py:602-604,611-612`). The
`store\_true` flags default `False`, so absent flags reproduce legacy/served behaviour on both
sides.

- Flags defined at `backend/eval/rally\_seg\_eval.py:712-723`.

```
---
```

4. DECISION RULE

- ****Headline metric:**** ****`tol\_f1` (tolF1)**** ? the R2 tolerance-match F1 at the ****SERVED****
operating point. This is the ***only*** headline metric named in the §8 Q2 / §9 decision rule.

- ****Promote vs reject (verbatim intent):**** base = SERVED flag-OFF, candidate = SERVED flag-ON;
the run "diffs tolF1 at the ****SERVED**** operating point." ? ****Promote**** by flipping
`IndexingConfig.inactivity\_temporal\_density` to `True` (one-line PR) ****only if `tol\_f1` does not
regress**** at SERVED. §9 restates: "flip the flag default-ON if no regression."

- ****`served\_regress\_eps`:**** ****Not named in the plan doc**** ? the doc's gate is the qualitative
"tolF1 does not regress." Grounded in code (not the doc), the promotion gate's tolerance is
`served\_regress\_eps: float = 0.0` at `backend/eval/promotion.py:126` (and `eps=0.0` in
`served\_regresses`, `backend/eval/promotion.py:42`), meaning ****"0.0 = no served-mean drop
allowed"***** (docstring line 147). Use `0.0` as the concrete bar unless the owner overrides it.

- ****Secondary metrics to inspect (harness outputs, NOT promote/reject gates):**** The plan doc
elevates none of these; they are diagnostics from the eval harness:

```

- **`|?end|`** ? per-video rally-end error (`rally_seg_eval` per-video table, `~line 523`);
the whole point of finding B is shrinking inflated ends.
- **Paired sign-test + CI / per-tier `delta`** ? module docstring line 13; sign-test on both
tIoU-F1 and R2 `tol_f1` (`~line 358`); aggregate CI via `_ci(a['tol_f1'])` (`~line 534`).
- **`tolP`/`tolR`/`tol_precision`** (tolerance precision/recall) ? note: a metric literally
called "rally-end precision" does not exist in the harness.
- **`promote_if_served_safe` gate** (`backend/eval/promotion.py:117-224`, the honest served-
safety gate, defaults `min_n=5`, `p_threshold=0.05`, `served_regress_eps=0.0`, `n_boot=2000`,
`seed=0`):
  1. `base = should_promote_for_user(served_a, served_b, ?)` ? honest gate on SERVED scores
(min-N floor + ?F1 CI excludes 0 + sign test).
  2. `served = served_regresses(served_a, served_b, eps=served_regress_eps, ?)`; `served_safe =
not served["regresses"]`. (`served_regresses` ? True when served mean ?F1 < ?eps or the
bootstrap CI lies entirely below ?eps.)
  3. Non-structural compose: if `would_promote` and not `served_safe` ? `promote=False`,
vetoed=True; elif `would_promote` ? `promote=True`; else `promote=False`. The served guard
can only ever remove a promotion, never add one.

> Decision boils down to: promote iff `tol_f1` does not regress at SERVED
(`served_regress_eps = 0.0`). Inspect `|?end|`, the paired sign-test, and the CI delta to
*understand* the result ? but they are not the gate.

```

5. ON PROMOTE (tol_f1 does not regress)

```

1. Exact one-line code change ? flip the default in `backend/config/models.py:199`:
`diff
- inactivity_temporal_density: bool = False
+ inactivity_temporal_density: bool = True
...

This is the only code change. It propagates automatically: config ? `trajectory_hybrid`
segmenter ? eval `windowingPreset` (`backend/eval/windowing.py:137-138,192-193` read it from
`IndexingConfig`) ? `tracknet_runner.trim_inactive_tail`.
2. Test update. Tests already cover both flag states (P0b shipped unit-tested ON and
OFF). After flipping the default, update any test asserting the default-OFF behaviour, and
consider updating the doc-comment block at `backend/config/models.py:193-198` / the preset doc-
comment at `backend/eval/windowing.py:74-76` to reflect the new default. Re-run the full suite
(see S7).
3. Doc updates:
- S7 tracker: P2b Status ?? ? ?, PR `?` ? `promotion PR link` (S7 is the source of
truth).
- S9 Definition-of-Done: check the box `[ ]` P2 promotion: golden-set A/B at SERVED
windowing, then flip the flag default-ON if no regression.
- Update the doc header "Last updated" date; when all rows land, advance the Status
header along DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED.
- Reindex inbound refs: `docs/README.md` (where S7 is indexed), and the
`RALLY_DETECTION_QUALITY_REPORT.md` cluster.
- DOC_STATUS archive protocol (only when the whole R-series/quality cluster reaches
terminal state ? do not archive yet per NEXT_STEPS): run the `docs/DOC_STATUS.md` S2 due-
diligence checklist before any `git mv` to `docs/archives/`: (1) verify claims vs code, tag
`[verified]`; (2) flip Status ? DELIVERED/DONE/REJECTED + completion note (what shipped, PRs,
paths); (3) update all inbound refs (`docs/README.md`, `CLAUDE.md`, `docs/NEXT_STEPS.md`, cross-
doc links); (4) keep ADR forward/back lineage; (5) `git mv` the whole file to
`docs/archives/past_projects/` (terminal-state only) and note relative links become historical;
(6) record the move in DOC_STATUS S3e + memory current-state.

```

6. ON REJECT (tol_f1 regresses at SERVED)

```

- No code change. Leave `inactivity_temporal_density` default-OFF
(`backend/config/models.py:199` stays `False`). P0b already shipped flag-OFF, so the merged
state is a zero-regression no-op in production ? rejecting requires nothing in code.

```

- **Record the numbers.** Capture the A/B output (``tol_f1`` base vs candidate, `?tolF1 + CI`, paired sign-test, ``|?end|``) in the doc so the rejection is auditable.
- **\$7 tracker:** mark P2b as **REJECTED** with a note (or keep ? with a rejection note), PR remains ``?``. Per CLAUDE.md, the flag stays gated default-OFF; the fix remains available behind the flag for re-measurement if the corpus grows.

7. FULL VALIDATION

> Tier-1 = OFFLINE ONLY (no GPU/real-video). Tier-2 (real video + full GPU/WSL/WASB byte-parity + golden corpus) is owner-gated on the self-hosted GPU runner ? exactly the box where P2b is run, so these gates **unlock** during this work.

```
**Offline suite (always):**
```bash
python run_all_tests.py # full suite, quiet (-q); exit code = pytest's
```

- `pytest.ini` sets `testpaths = tests` (scratch/ never collected). Fully mocked: no GPU/network/WSL. The runner probes `obsreport` and prints availability (reporting tests self-skip via `pytest.importorskip('obsreport')`).
- Expected count: documented latest 1492/1492 green
(`docs/CODE_CLEANUP_AUDIT_2026-06-24.md`); a fully-equipped local checkout collects ~1513 (parametrize expansion + optional deps present). CI lane reports fewer (obsreport/golden absent).

**Now-unlockable GPU / real-video / golden / WASB gates (run on the owner box where the golden corpus is present):**
- `tests/test_served_gate_a_regression.py` ? runs when the manifest at env `SERVED_GATE_A_MANIFEST` (default `<repo>/output/human_lovo_manifest.json`) yields ?6 specs with present trajectory+labels files; also gated by `pytest.importorskip('cv2')`. (Skips on a fresh worktree.)
- `tests/test_ablation.py` ? `skipif(not _golden_present())`; needs ?6 `output/*_golden_features.json` with the fusion schema; the LITMUS test reproduces Gate-A (`?f1 ~ +0.074`, 6/6, p ~ 0.031). Also `importorskip` `reportlab + pypdf` for PDF report tests.
- `tests/test_native_wasb_runner.py` ? native WASB contract (subprocess mocked). The byte-for-byte GPU parity diff is the owner-box gate, not a pytest test; pytest only asserts `wasb_infer.py` byte-identity (`test_copy_infer_script_copies_byte_identical`).
- `tests/test_golden_fixtures.py` / `tests/test_golden_real_fixtures.py` ? committed de-attributed fixtures; do run in CI via `python -m backend.eval.golden_fixtures --check`.
- Set the relevant env drivers as needed: `RALLY_DETECTOR_IMPL=native|stub`, `WASB_WEIGHTS/WASB_DEVICE/WASB_PYTHON/WASB_REPO_DIR/WASB_SCRATCH`, `SERVED_GATE_A_MANIFEST`.
```

```
**Lint / types / coverage (must stay green):**
```bash
ruff check . --output-format github
mypy backend training
python run_all_tests.py --cov=backend --cov=training --cov-report=term --cov-fail-under=70
```

- ruff (`ruff.toml`): lint-only (no `ruff format`), `target_version = py38`, rules = defaults + `W,B,I`; `extend-exclude = ['scratch']`; `E402` per-file-ignored for `backend/main.py` and two `scripts/*`.
- mypy (`mypy.ini`): lenient green baseline; `ignore_missing_imports = True`, `explicit_package_bases = True`; quarantined-ignore modules: `backend.main`, `backend.worker.__main__`, `backend.pipeline.segmenters.{yolo_hybrid,gemini,base}`. Ratchet down (remove a quarantine entry) only when you touch that module.
- coverage floor: `--cov-fail-under=70` (calibrated against CI's measured ~72%). Ratchet up as coverage grows; never lower without an owner decision. Omits in `coveragerc` (legacy `motion.py`, concrete validators).
```

8. GUARDRAILS (CLAUDE.md)

- **Owner-gated work:** No platform/owned-model implementation has started ? do **not** start without explicit owner approval. Hand the heaviest machine-side checks (real video, full GPU/WSL/WASB suite) to the owner who holds the golden corpus + labelled videos; verify pure-logic paths locally. P2b promotion itself is owner-gated.
- **Commercial-clean licensing** (ratified 2026-06-09): every dependency ? **code, data, AND weights** ? must be **MIT / Apache-2.0 / BSD only**. Enforced by the CI license-scan lane (``pip-licenses``, fails on ``GPL|SSPL``). Owned-model base: Qwen3-VL (Apache-2.0) primary, InternVL3 (MIT) fallback; **reject Gemma 3**; Gemma-4 is AMBER/bench-only. (Commercial-ship blocker: C3 WASB checkpoint provenance ? out of P2b scope but noted.)
- **Paid LLMs** (Gemini/OpenAI/Anthropic) = inference/serving/fallback **ONLY** ? never a training-data source. Exclude minors; per-user training data needs explicit opt-in.
- **Compute = GCP only** (ADR-013): GPU training + serving on GCP (Cloud Run + GPU per ADR-012). Don't re-evaluate DigitalOcean/Paperspace without a new ADR.
- **Scope:** v1 = rally + highlight + history + correction loop; gait/biometrics strictly **FUTURE** (GDPR Art. 9).
- **Branch / PR discipline:**
 - Branch off ``master`` (origin/master). **ONE PR per work item; NO PR stacking** ? one small PR per \$7 row.
 - **Do NOT open a PR unless asked.** Never push to the default branch directly; descriptive commit messages.
 - Tests: ``python run_all_tests.py`` (offline, fully mocked). Python **3.10+** (trackers needs 3.10); ffmpeg required.
 - **Archive policy:** only move terminal-state (DONE/REJECTED) work to ``docs/archives/``; honestly update the doc first (verify ? flip status ? update inbound refs ? keep lineage ? ``git mv``).

9. OPEN QUESTIONS / UNKNOWNNS

- **Concrete golden paths not in the plan doc.** ``<GOLDEN_MANIFEST>`` and ``<GOLDEN_FEATURES_DIR>`` are placeholders in §8 Q2 with no concrete path. Best available leads (from code/tests, not the plan): manifest default ``<repo>/output/human_lovo_manifest.json`` (env ``SERVED_GATE_A_MANIFEST``); features as ``output/*_golden_features.json``. **Owner must confirm the actual corpus layout.**
- **Exact feature-gen command is unspecified.** The plan doc pins only the ``rally_seg_eval`` A/B command, not the upstream GPU/WSL trajectory-feature extraction command. ``python -m backend.eval.golden_fixtures --check`` is the committed *sanity* gate, not the extraction step; the real extraction is the WSL-quarantined ``backend/pipeline/detectors/wasb_infer.py`` path. **Owner to supply the exact extraction invocation.**
- **No numeric regression epsilon/p-value/CI rule in the plan doc.** The gate is the qualitative "tolF1 does not regress." The only concrete value is ``served_regress_eps = 0.0`` from ``backend/eval/promotion.py:126`` (code, not doc). Confirm with the owner whether ``0.0`` (no drop allowed) is the bar or whether a tolerance/p-value applies via ``promote_if_served_safe``.
- **Secondary metrics are not gates.** ``|?end|``, paired sign-test, and CI delta exist in the harness but are **not** elevated by the plan into the promote/reject rule; "rally-end precision" as a named metric does not exist (harness reports ``tolP/tolR/tol_precision``).
- **Doc inconsistencies to reconcile during the doc update:** (a) §7 PR **#394** vs header **#393** for P0a/P0b; (b) §7 P1 ? Done vs §9 unchecked ``rally_cvat.py`` warning-log box. Resolve both when touching the tracker.
- **Ship-gate target still owner-unset.** The formal ship-gate number (NEXT_STEPS #7 / #172, suggested shape ``LOVO F1@tIoU0.5 ? 0.70`` multi-court, $p < 0.05$ vs heuristic) is unrelated to the P2b tolF1-no-regression gate but is the broader blocker; do not conflate.
- **Cluster not archive-ready.** Per NEXT_STEPS/DOC_STATUS, the R-series/quality doc cluster must **not** be archived until R2 tolF1 is promoted-or-rejected as the gate and the §7.1 single quality number is attributed ? i.e., P2b's outcome is itself a prerequisite for archival.